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PHARMACOGNOSY - II

UNIT - 1st METABOLIC PATHWAYS IN

HIGHER PLANTS & THEIR DETERMINATION

• Syllabus :-

a) Brief study of basic metabolic pathways and formation of different secondary metabolites through these pathways -

- 1) Shikimic Acid Pathway
- 2) Acetate Pathway
- 3) Amino acid Pathway

b) Study of utilization of radioactive isotopes in the investigation of biogranitic studies.

• Higher plants → (Vascular plants)

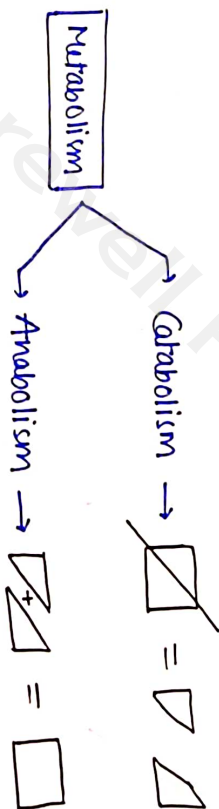
- Completely fully developed plants
- Distributed resources to others.

• Metabolism →

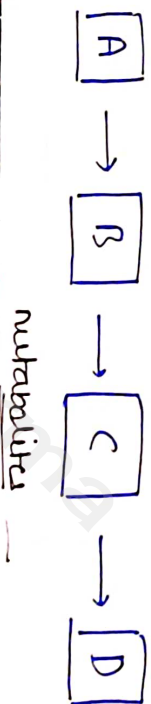
The chemical process in plants or animals that changes food into energy and help them grow.

[OR]

(Conversion of one chemical entity to another inside the living beings (plants or animals) with the help of enzyme.



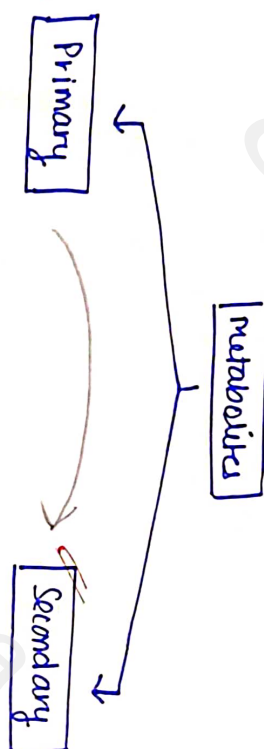
• The products and intermediate which forms during metabolism is known as metabolites.



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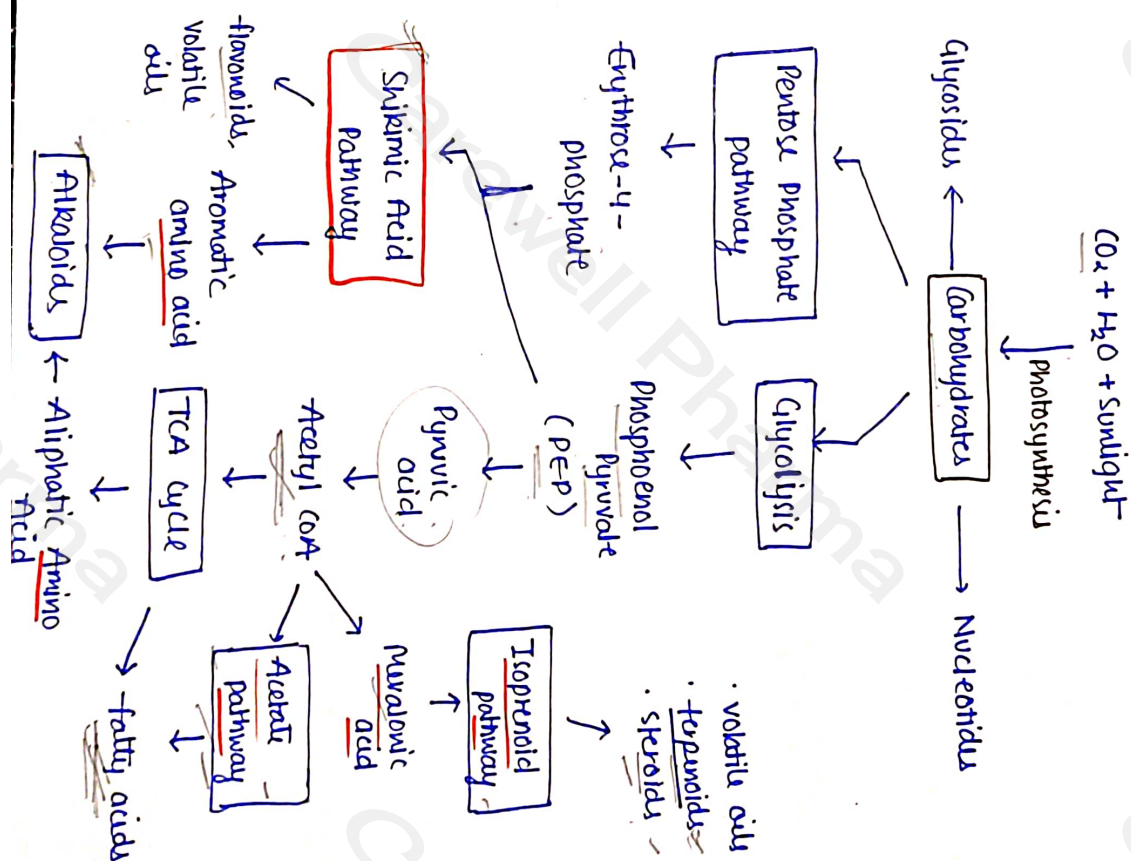
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- Metabolites are of two types:-



- those metabolites which directly involved in growth, development and reproduction of plants.
- example - Proteins, Carbohydrates, lipids, vitamins, Amino acids, hormones etc--
- those metabolites which biosynthesized from a primary metabolites & does not involved in growth, development.
- Act as a defence system for plants
- eg.- Alkaloids, steroids, essential oils, glycosides etc--
- Metabolic pathway → It is a linked series of chemical reactions occurring within a cell.

- Basic Metabolic pathway:-



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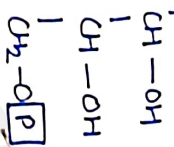
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SHIKIMIC ACID PATHWAY

This pathway used in the biosynthesis of Aromatic amino acid which further used in the synthesis of secondary metabolites (alkaloids etc--)

from Pentose

Phosphate pathway

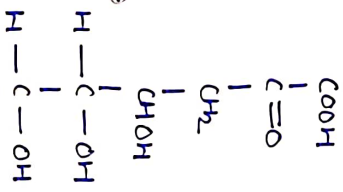
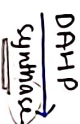


erythrose-4-phosphate

from glycolysis

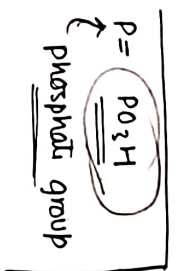


Phosphoenolpyruvate (PEP)

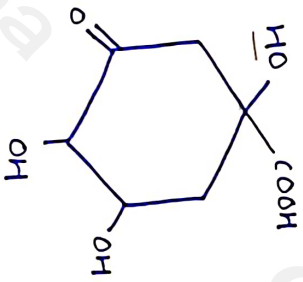
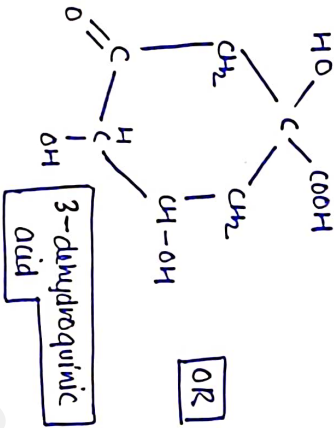


2-keto-3-Deoxy-7-phospho D-gluco heptonic acid

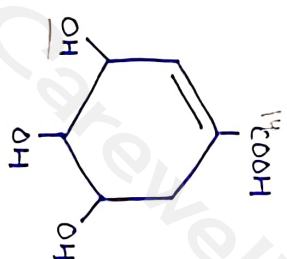
3-Deoxy-D-arabino-heptulosonic acid-7-phosphate (DAHP)



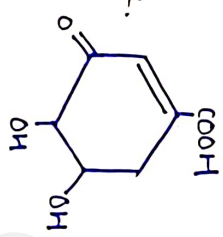
Hydration 3-dehydroquinate synthase



3-dehydroquinoic acid

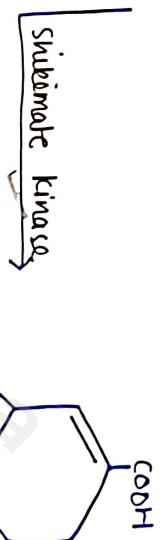


Shikimate dehydrogenase

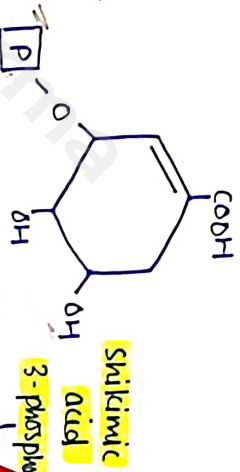


3-dehydro Shikimic acid

Shikimic acid



Shikimate kinase



Shikimic acid 3-phosphate

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- Shikimic acid pathway starts with precursors, Fructose 4-phosphate and Phosphoenol pyruvate
- which on aldol condensation yield DHAP - 3-deoxy-D-arabino-heptulosonic acid-7-phosphate with the help of enzyme 8p DHAP synthase.
- The enzyme 3-dehydroquinate synthase, catalysing the cyclization of DHAP to 3-dehydroquinate acid.
- Now, 3-dehydroquinate acid give 3-dehydroshikimic acid on dehydration which further give shikimic acid with the help of enzyme shikimic acid dehydrogenase.
- Now, shikimic acid through phosphorylation with the help of enzyme shikimate kinase give 3-enol pyruvyl shikimic acid 5-phosphate,
- which further convert into chorismic acid with the help of enzyme chorismate synthase.
- The formation of chorismic acid is an important branch point in the shikimic acid pathway as this compound undergoes different type of conversion.
- Chorismic acid with the help of enzyme anthranilic acid synthase convert into anthranilic acid which act as a precursor for tryptophan (imp. amino acid)
- Chorismic acid gives prephenic acid with the help of enzyme chorismate mutase.
- Now this prephenic acid give phenylpyruvic acid on aromatization by enzyme prephenate dehydratase which further on transamination give phenylalanine.
- Also, prephenic acid give 4-hydroxyphenylpyruvic acid on aromatization by enzyme prephenate dehydratase which further on transamination give tyrosine.
- Tyrosine and phenylalanine both act as a precursor for the biosynthesis of phenylpropanoids → produce flavonoids, liquin etc.

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ACETATE PATHWAYS

Also known as Acetate mevalonate pathway

Isoprenoid pathway mevalonic acid

Terpenoid pathway pathway

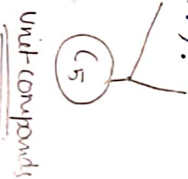
HMG Co-A Reductase pathway

This pathway is used in the biosynthesis of essential secondary metabolites such as isoprenoids (terpenoids) and steroids

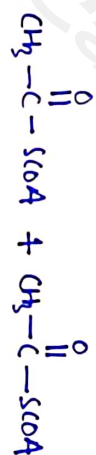
It is an essential metabolic pathway present in eukaryotes, archaea and some bacteria

This pathway produces two 5-Carbon building blocks called isopentenyl pyrophosphate (IPP) and Dimethyl allyl pyrophosphate (DMAPP).

It also used in synthesis of fatty acid and Polysketides through threate malonate pathway

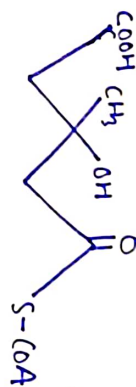


from pyruvic acid (glycolysis)



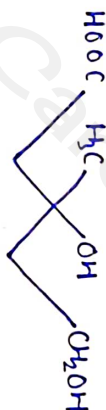
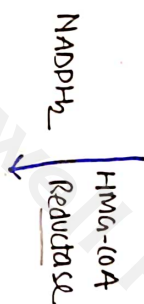
Acetyl CoA + Acetyl CoA

Claisen condensation

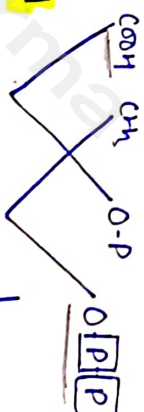
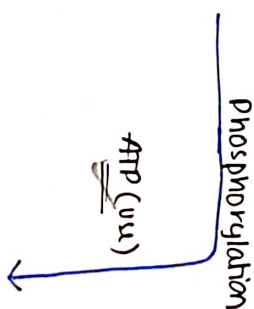


3-Hydroxy-3-methyl glutaryl-CoA (HMG-CoA)

(HMG-CoA)



Mevalonic acid



5-pyrophosphate-3-

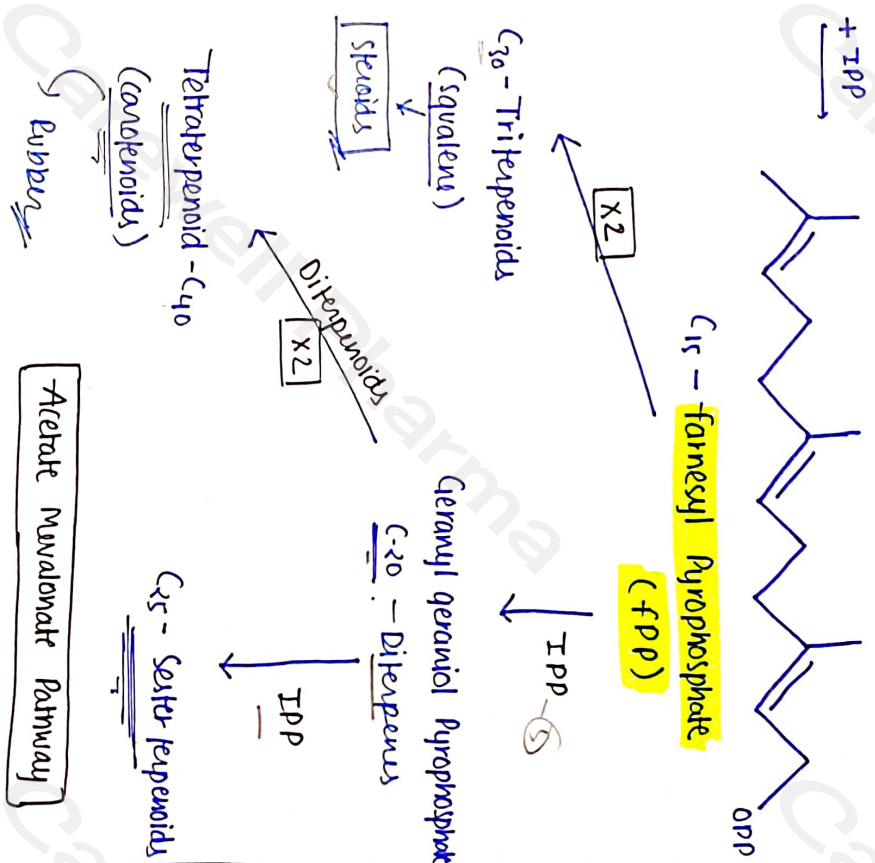
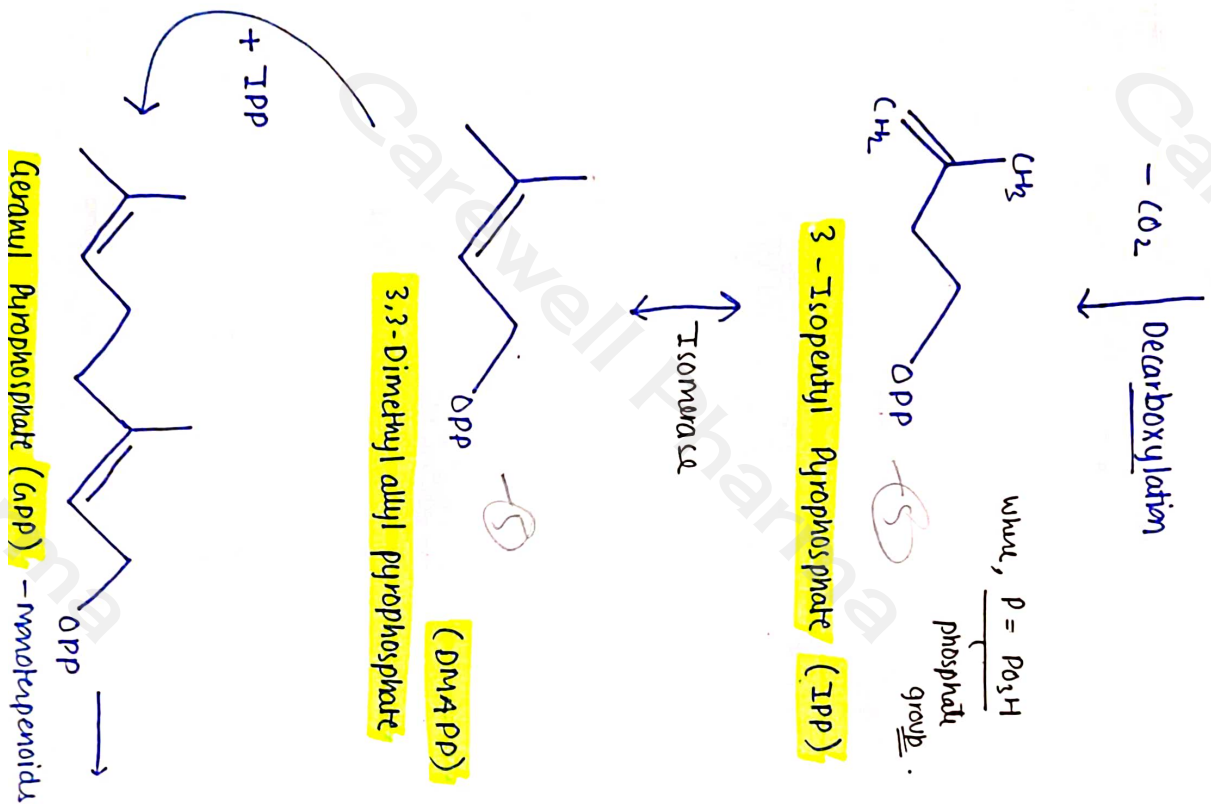
Phospho mevalonic acid

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- The precursor for this pathway is Acetyl co-A, which is the end product of glycolysis, which is produced from pyruvic acid.

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- Now, first two molecules of Acetyl CoA forms acetoacetyl CoA through chain condensation.
- which further on aldol addition gives β -Hydroxy β -methylglutaryl-CoA (HMG-CoA).
- Now, HMG-CoA on reduction give mevalonic acid, which is the main precursor for biosynthesis of terpenoids.
- Now, Mevalonic acid on ATP mediated phosphorylation give diphosphate of mevalonic acid
- which further on decarboxylation give 1st isoprenoid unit - Isopentenyl pyrophosphate (IPP).
- which further convert into 2nd isoprenoid unit - Dimethyl allyl pyrophosphate (DMAPP) with the help of isomerase enzyme.
- Now, Addition of IPP unit with DMAPP with the help of enzyme prenyl transferase yield C_{10} unit Geranyl pyrophosphate (GPP), which is the precursor for the synthesis of monoterpenes.
- Now, Addition of another IPP unit with GPP give farnesyl pyrophosphate (FPP)- C_{15} which is the precursor for synthesis of sesquiterpene.
- further addition of IPP unit with FPP give C_{20} geranyl geraniol pyrophosphate (GGPP) to produce diterpenes.
- further addition of IPP unit with GGPP give C_{25} geranyl farnesyl pyrophosphate (GFPP) also called sesterpenes.
- Now, Addition of two FPP (C_{15}) unit give C_{30} unit triterpenes which also used to produce squalene to steroids.

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- Similarly, addition of two GPP unit give C_{40} unit compound i.e. tetraterpenoids which further used to produce carotenoids
- Also further on addition of more C unit give rubber etc--

Carbon unit	Name
C_{10}	Monoterpene
C_{15}	Sesquiterpene
C_{20}	Diterpene
C_{25}	Sesterpene
C_{30}	Triterpene
C_{40}	Tetraterpene
C_5	IPP, DMAPP

- This acetate mevalonate pathway produce various terpenoids and steroids which has many uses in human body

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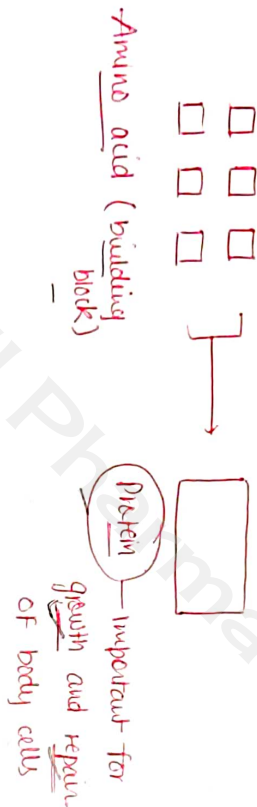
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AMINO ACID PATHWAY

Amino acids are molecules that combine to form proteins. Amino acid and proteins are the building blocks of life.



- There are total 20 amino acid which needs in our body
- Plants and bacteria can synthesize all 20 of the amino acids but humans cannot synthesize 9-10 of them.
- Those amino acids which cannot synthesized in our body are known as essential amino acids.

- Those amino acids which can synthesized in our body are known as non-essential amino acids.

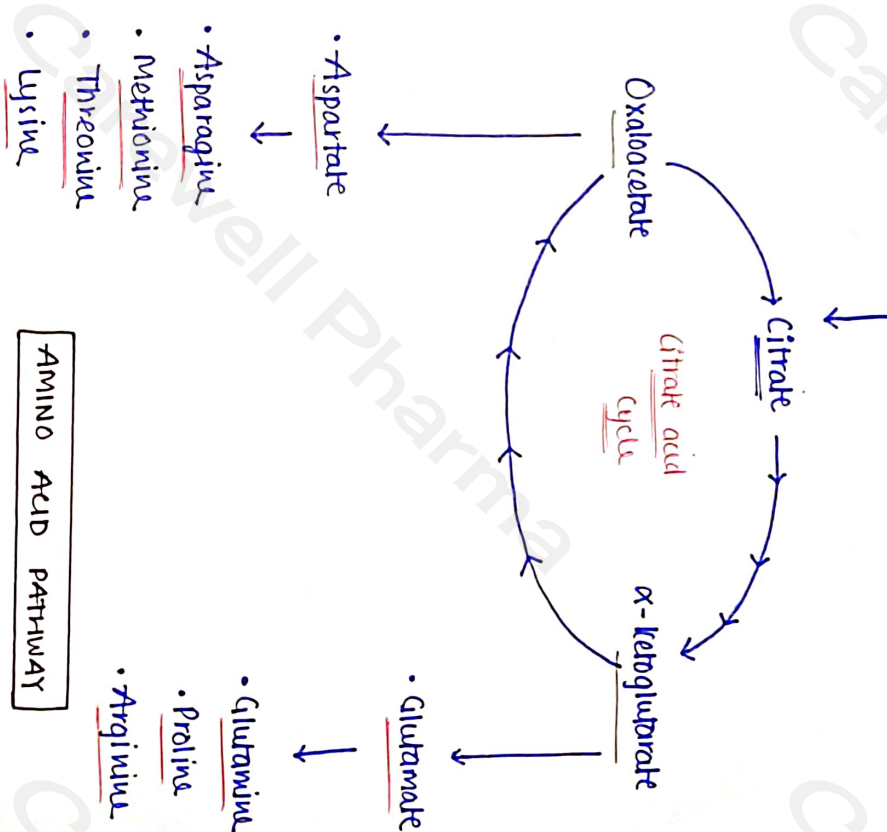
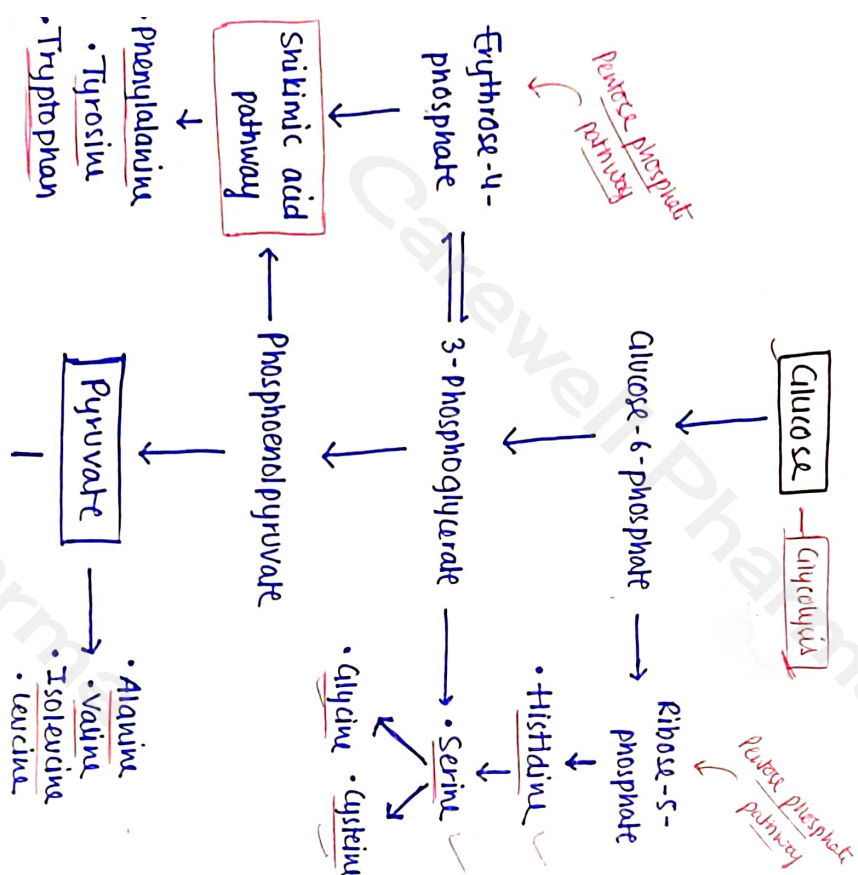
Essential Amino acids	Non-essential amino acids
Arginine*	Alanine
Histidine	Asparagine
Isoleucine	Aspartate
Leucine	Cysteine*
Lysine	Glutamate
Methionine	Glutamine*
Phenylalanine	Glycine
Threonine	Proline*
Tryptophan	Serine
Valine	Tyrosine

* Can be synthesized in our body.
Conditionally* | Arginine can be Non-essential in some conditions.

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- Now, these essential amino acids must come from our diets
- The non-essential amino acids are synthesized by simple pathways, whereas biosynthesis of the essential amino acid are complex.



- All amino acids are derived from intermediate in glycolysis, citric acid cycle and pentose phosphate pathway.

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Study of utilization of radioactive isotopes in the investigation of Biogenetic studies

• Isotopes → These are those element which have same chemical properties or same atomic number but different atomic mass.



– Radioactive Isotopes → These are unstable and emit radiations.

• The phenomena of emitting radiation is called radioactivity and such isotopes are called radioactive isotopes. eg. ^{14}C , ^3H , ^{32}P , ^{131}I

– Biogenetic studies → study of synthesis of primary and secondary metabolites in living beings (plants).

• Various intermediate and steps are involved in biosynthetic pathway in plants can be investigated by following technique

- i) Tracer technique
- ii) Grafting method
- iii) Use of isolated organ and tissue
- iv) Use of mutant strains
- v) Enzymatic studies.

• Now, In tracer technique method radioactive isotopes are used for the investigation of biogenetic studies.

→ Tracer technique ⇒

It can be defined as technique which utilizes a labelled compound to find out or to trace the different intermediate and various steps in biosynthetic pathways in plants at given rate + time

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- for different studies, different tracers can be used :-

$$\left. \begin{array}{l} \text{protein, Alkaloids and} \\ \text{Amino acids} \end{array} \right\} \text{Nitrogen}$$

$$\begin{array}{l} \text{Glycosides} \rightarrow \text{O, N, S \& C} \\ \text{Terpenoids} \rightarrow \text{O atoms} \end{array}$$
- steps involved in tracer technique
- i) Preparation of labelled compound
- ii) Incorporation of labelled compound
- iii) Separation and isolation of labelled compound
- iv) Determination of nature of metabolites in various biochemical fractions.
- i) Preparation of labelled compounds :-
 firstly we have to prepare radioactive labelled compounds.

- labelled compounds may be prepared by use of radioactive isotopes.
 eg $\left(\text{}^{14}\text{C}, \text{}^3\text{H}, \text{}^{32}\text{P}, \text{}^{131}\text{I} \right)$
- Also sometimes stable isotopes - $\text{}^2\text{H}, \text{}^{15}\text{N}, \text{}^{12}\text{C}, \text{}^{18}\text{O}$
- Mostly Carbon and hydrogen are used in biological investigation.
- Choose those isotopes which have long half-life.
- These radioactive isotopes can be prepared through bombarding
- ii) Incorporation of labelled compound
 - After preparation of radioactive isotopes, we have to introduce (administer) into plants.
 - following method for this :-
 - i) Root feeding $\rightarrow$ In case, roots are biosynthetic sites eg. Tobacco.

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ii) Stem feeding → In this, labelled compounds are administered through the cut end of stem immersed in a solution.

iii) Direct injection → labelled compound's solution is injected in plants with hollow stem. eg. Opium etc.

iv) Spraying technique → labelled compounds have been observed after being sprayed on leaves. eg. steroids

• Now, labelled compound administered to the plant from any technique

— After administration, labelled compound becomes a part of metabolic pathway and undergoes reaction characteristics

iii) Separation and isolation of labelled compounds after radioactive tracers are used in biogeochemical studies, its separation and isolation is essential.

— different methods are used depending on nature of drugs and its source.

eg. • Soft tissue → Infusion, incubation

• Alkaloids, glycosides, flavonoids → process of extracting chemical compound from plant material.
 tissue is transformed

slightly polar solvent into a suspension of extract cells.

iv) Determination of nature of metabolites in various biochemical fractions →

— Depending upon the nature of isotopes, various instruments are used to determine the chemical nature of intermediate and final product for radioactive isotopes.

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eg.

- Geiger - Muller Counter
- Scintillation
- Ionization chamber
- Mass spectrophotometer
- NMR spectrophotometer
- Autoradiography



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